

Instructions for Administration:

Birth to < 1 Month

Note Ensure to wash your hands before handling the child and to handle the child gently. For neonates and young infants (<3 months) offer a little bit of time after testing each item, to allow the infant to settle.

Definitions

Plane: In reference to items requiring holding a body part on the same plane as another, this refers to being level to.

Prone Position: The prone position refers to lying horizontally with the face and torso facing down. The infant lies with knees drawn up under the abdomen with pelvis high. Head is turned to one side.

Supine Position: The supine position refers to lying horizontally with the face and torso facing up.

Ventral Suspension: The baby is placed in prone position and suspended in the air by a hand placed under the chest. The head hangs down with some flexion of elbows and knees.

1.0 Primitive Reflexes

1.0.1	Reflex: Rooting	Age: Birth - < 1 month
Materials	Padded Surface, Hygienic Gloves	
Overview	The rooting reflex is well established by 32 weeks gestation. It is demonstrable in the newborn period and then fades during the early months.	
Instructions	Place the infant in supine position, head in the midline. Touch one corner of the mouth and stroke laterally. Do this on both sides. 	
Response	Turning of the head in the direction of the stimulus, simultaneous opening of the mouth and extension of the tongue.	
Grading	Present Weak None	

1.0.2	Reflex: Sucking	Age: Birth - <1 month
Materials	Padded Surface, Hygienic Gloves (or have caregiver test reflex)	
Overview	Sucking and swallowing are coordinated sufficiently for oral feeding as early as 28 weeks gestation. By 32 to 34 weeks gestation, the typical newborn infant can maintain a consistent synchronous action to enable oral feeding. Maturation continues and linkage of breathing, sucking, and swallowing is not achieved fully until 37 weeks gestation age or more.	
Instructions	This reflex is assessed by placing the little finger (pad towards palate) in the mouth of the infant. 	
Response	The rhythm and strength of the sucking response is noted.	
Grading	Present Weak None	

1.0.3	Reflex: Palmar Grasp	Age: Birth - <1 month
Materials	Padded Surface	
Overview	Palmar grasp is present clearly at 28 weeks gestation. It is strong at 32 weeks gestation and strong enough and associated with extension of upper limbs (when lifted from bed) at 37 weeks gestation. The palmar and plantar grasp reflexes are easily demonstrable in the neonate but then fade rapidly and are seldom seen after 4 or 5 months of age.	
Instructions	Place the infant in supine position with the head in the midline. Each side is tested, separately or together, by placing your index finger in the palm of the infant's hand. 	
Response	Strong flexion of the fingers.	
Grading	Present Weak None	

1.0.4	Reflex: Moro Reflex	Age: Birth - <1 month
Materials	None	
Overview	The Moro reflex is a particularly useful diagnostic reflex. It is present consistently and is so easily elicitable in the newborn period that variation from the fully normal response at this time is a reliable indication of probable abnormality. It is not normally elicitable after 6 months of age, when its persistence then indicates abnormality. Asymmetry of the Moro reflexes can be produced by lower motor neuron lesions, whereas upper motor neuron lesions do not produce asymmetry. In infants with impaired consciousness or with basal ganglia lesions, it can be quite difficult to elicit the Moro reflex. Therefore, the threshold at which the reflex can be elicited should also be noted.	
Instructions	When the infant's condition permits, use the 'head drop' method. The infant is held suspended in supine position, raised at 45 degrees with one hand supporting the back and the other supporting the head. The infant's arms should be over the chest. The head is held in a midline position then dropped back 10 degrees.	
Response	Consists of opening of hands and extension and abduction of the upper extremities followed by an anterior flexion of the upper extremities. The response is based on the observation of the arm movements only. The test may be repeated 2 to 3 times to get a detailed observation.	
Grading	Present Weak None	

1.0.5	Reflex: Placing	Age: Birth - <1 month
Materials	Table or similar surface	
Overview	This reflex is readily demonstrable in the newborn. Persistent failure to elicit it at this stage is thought to indicate neurological abnormality. It fades rapidly in the early months.	
Instructions	The infant is held upright in a vertical position by placing one or both hands around their chest under the arms, holding their neck with fingers. Stroke the dorsum of the infant's foot with the edge of the table or another surface. Test each side separately. 	
Response	Stimulation of the dorsum of the foot of the neonate produces flexion of the same leg, followed by placement of the sole of the foot on the surface.	
Grading	Present Weak None	

1.0.6	Reflex: Stepping Reflex	Age: Birth - <1 month
Materials	Table or similar surface	
Overview	'Stepping' is not true walking as there is no trunk support or pelvic stability. The stepping reflex should disappear by 6 months of age.	
Instructions	The infant is held upright in a vertical position by placing one or both hands around their chest under the arms, holding their neck with fingers. Have the soles of the infant's feet touching the examination surface. 	
Response	Pressure upon the soles of the feet first causes flexion then extension of the legs. As these occur on alternate sides, an impression of automatic stepping is created.	
Grading	Present Weak None	

1.1 Gross Motor

Notes If the child does not perform the gross motor item for their age, but performs a more advanced gross motor item, then the item specific for their age is considered normal.

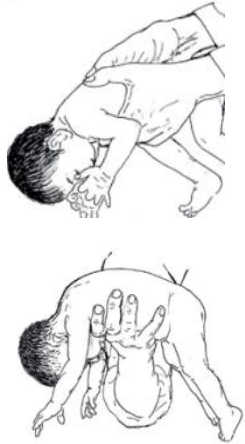
The first step towards the development of locomotion is the development of head control, that is, the ability to support the head in all positions of the body. The steps in this are observed in three situations: in ventral suspension, in prone position, and in supine position.

1.1.1	Posture in supine position	Age: Birth - <1 month
Materials	Padded Surface	
Overview	Most of the neurological items assessing posture and tone are age dependent, reflecting the increase in flexor tone in the limbs and increase in axial tone with maturation.	
Instructions	Posture is assessed with the infant in supine position and head in the midline, after gently uncovering the infant and taking off or loosening the nappy with as little disturbance as possible.	
Response	In supine position, the infant assumes a flexed posture of all four limbs. There is often a preference for position of the head towards the right side (right 79%, left 19%, midline 2%). arms flexed Abnormal posture: opisthotonus arms flexed, legs extended	
Grading	Limbs well flexed Limbs slightly flexed Limbs persistently extended/Arms strongly flexed with legs extended/Opisthotonus	

1.1.2	Writhing movements of four limbs in supine position	Age: Birth - <1 month
Materials	Padded Surface	
Instructions	Observe the infant in supine position. Note spontaneous movements for a couple of minutes, preferably while the infant is awake and quiet. Observe both the quality and quantity of movements. Evaluate the patterns and the sequence of movements of the whole body, rather than concentrating on single segments, such as arms and legs.	
Response	The movements should be fluent and alternating.	
Grading	Present Sluggish None / Asymmetric / Continues exaggerated	

1.1.3	Lifts chin up momentarily when in prone position	Age: Birth - <1 month
Materials	Padded Surface	
Overview	The first step towards the development of locomotion is the development of head control – that is, the ability to support the head in all positions of the body.	
Instructions	The infant is placed in prone position with the head in the midline, and the movements of the head are noted for about 30 seconds.	
Response	Lifts chin up momentarily when lying in prone position.	
Grading	Raises chin, rolls head over Feeble movement of head No response OR persistently raised head	



1.1.4	Fleeting tensing of neck muscles in ventral suspension	Age: Birth - <1 month
Materials	Padded Surface	
Instructions	From prone position, lift the infant horizontally with one or both hands under the chest so they are in ventral suspension, about 20-25cm from a surface. Note the infant's attempt to lift the head for about 30 seconds. 	
Response	The fleeting tensing of neck muscles.	
Grading	Present Feeble response No response OR persistently raised head	

1.1.5	Lifts head on pull to sitting	Age: Birth - <1 month
Materials	Padded Surface	
Instructions	Place the infant in supine position, grasp the shoulders and pull the infant towards a sitting position. Note the posture of the head when the shoulders are elevated to about 45 degrees. 	
Response	Tries to lift head but it drops back.	
Grading	Shows effort, but drops back Feeble Response No Effort	

1.2 Fine Motor Item

Note If the child does not perform the fine motor item for their age, but performs a more advanced fine motor item, then the item specific for their age is considered normal.

1.2.1	Hand posture		Age: Birth - 1 month
Materials	None		
Instructions	Observe the infant when awake and alert, and at rest.		
Response	Intermittent fisting, with thumbs in or out of the palms.		
Grading	Present Atypical Hand Posture, either: a) continuous fisting OR thumb adduction b) index finger flexion, thumb opposition		
Reference	Typical Hand Posture 	Atypical Hand Posture A) <i>Thumb Adduction</i> 	Atypical Hand Posture B) <i>Opposition of index finger and thumb (other three fingers extended)</i> 

1.3 Vision Items

Note Administer all three items, but for this age group only the **best response** to any one of the three visual stimuli is considered for scoring.

1.3.1	Response to light (glowing torch beam/diffuse light)	Age: Birth - <1 month
Materials	Torch, Room with a window, Padded Surface	
Instructions	Ensure the room is darkened. Position the infant in supine position on a flat surface or along the caregiver's lap on extended arms. Use the small torch from the kit to briefly focus light on the infant's eyes. If using a different torch that emits bright light, cover it with a yellow cloth. This is because dim light is preferable to bright light for this task, and due to infant light sensitivity. For response to diffuse light, hold the infant in supine position on the examiner's extended arms and turn slowly from the relative shadow inside the room toward the light source from the window.	
Response	Blinks and/or regards torch light OR blinks/shifts eyes/turns towards diffuse light.	
Grading	Present Questionable None	



1.3.2	Response to face	Age: Birth - <1 month
Materials	None	
Instructions	Ensure the room is well-lit. The infant may be held on the examiner's lap or tested in a propped up supine position. With the examiner's face 15-25cm in front of the infant's face, move slowly in a horizontal and vertical arc. Avoid talking to the infant during this item.	
Response	Regards face.	
Grading	Present Questionable None	



1.3.3	Response to spinning woolly ball	Age: Birth - <1 month
Materials	Red Woolly Ball	
Instructions	Ensure the room is well-lit. The infant is tested in a propped up supine position and the stimulus is presented at a distance of 15-25 cm, starting in the midline and moving side-to-side in either direction, then up-and-down.	
Response	Regards the woolly ball.	
Grading	Present Questionable None	

1.4 Hearing Items

Note Soft whisper - 10 to 25 db; conversational speech - 25 to 55 db; crying baby - 55 to 65 db; rattle - 60 to 80db.

1.4.1	Response to clapper bell	Age: Birth - <1 month
1.4.2	Response to rattle	
1.4.3	Response to voice	
Materials	Clapper Bell, Rattle, Assistant, Padded Surface	
Instructions	The infant is held in a propped up supine position on a padded surface, or on the caregiver's lap, with the head supported in the midline. The auditory stimulus is: (a) presented at the ear level, (b) about 15 cm from the ear, (c) out of the line of vision. Test the right ear then the left ear. A trained assistant provides the stimulus, and the tester observes the response.	
		
		
Response	Infant will brighten/startle/eyes will widen/become still/show respiratory changes/blink/shift of eyes to sound. The best response to any one of the three sound sources is considered for scoring.	
Grading	Present Response to Louder Ring/Louder Rattle/Voice No response	

1.5 Expressive Language Item

Note This domain relies primarily on observation of infant vocalisations during the session:

- The examiner has to be vigilant about recording all infant vocalisations during the entire assessment so that none are missed.
- Interact and speak with the child to encourage vocalisations.
- If no vocalisations are made throughout the assessment, ask the caregiver if this is typical as it is not uncommon for young children to behave differently in new and unfamiliar situations. Document specific examples of vocalisations reported by the caregiver on the record form.

1.5.1	Loudness of cry	Age: Birth – 1 month
Materials	None	
Instructions	Crying in the newborn usually falls within a typical range for pitch and loudness.	
Response	Note the typicality of the loudness of the crying	
Grading	Typical/Within normal limit Poor Crying Sound No Crying Sound	

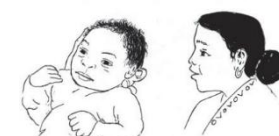
1.6 Cognition Items

Note Administer all three items, but for this age group only the **best response** to any one of the three stimuli is considered for scoring.

1.6.1	Degree of alertness, response to face	Age: Birth - <1 month
Materials	None	
Instructions	The infant may be held on the examiner's lap or tested in a propped up supine position. With the examiner's face 30-40cm in front of the infant's face, move slowly in a horizontal and vertical arc. Avoid talking to the infant during this item.	
Response	Observe if the infant alerts to the face, which may be expressed as a brightening of the expression and/or widening of the eyes.	
Grading	Present Questionable None	



1.6.2	Degree of alertness, response to voice	Age: Birth - <1 month
Materials	Assistant	
Instructions	The infant is tested in a propped up supine position or held in the caregiver's lap. The assistant speaks softly into one of the infant's ears outside of the infant's line of vision, about 15-30cm away from their face. The sounds of the voice may be varied in pitch and intensity to obtain the infant's attention and avoid habituation. Usually, a soft high-pitched voice is the most potent stimulus.	
Response	Observe if the infant alerts to the voice, which may be expressed as brightening of expression and/or widening of the eyes.	
Grading	Present Questionable None	



1.6.3	Degree of alertness, response to touch	Age: Birth - <1 month
Materials	None	
Instructions	Touch and stroke the infant gently over the arms and face. 	
Response	Observe if the infant alerts to the touch, which may be expressed as brightening of expression and/or widening of the eyes.	
Grading	Present Questionable None	

1.7 Behaviour Items

Note It is essential the examiner scores the Behaviour items based on their observations and caregiver report regarding sleep. Observe and record the child's behaviour according to the activities listed under this domain.

1.7.1	Cry	Age: Birth - <1 month
Materials	None	
Instructions	This item has been included to document the quantity and quality of the infant's cry at any time during the assessment. It may prove a useful index of atypicality when the cry is diminished or excessive or altered in character.	
Response	Note the following responses: (1) no cry at all even with a gentle stimuli (2) whimpering cry only (3) cries to gentle stimuli but normal pitch (normal pattern) (4) high-pitched cry and often continuous (this can suggest cerebral irritation which can indicate brain inflammation, or neurological disorders like epilepsy. If the child has epilepsy or seizure disorders, continuous crying can sometimes occur before, during, or after a seizure episode, especially in infants).	
Grading	Normal pattern Whimpering cry only No cry at all OR High-pitched, often continuous cry	

1.7.2	Consolability to crying	Age: Birth - <1 month
Materials	None	
Instructions	This measures the infant's ability to be consoled once they have reached a crying state, either spontaneously or by various manoeuvres (e.g., cooing, rocking, feeding, changing nappy) carried out by the caregiver	
Response	Note the infant's response to manoeuvres of the caregiver.	
Grading	Good With difficulty Unconsolable	

1.7.3	Irritability (crying to handling)	Age: Birth - <1 month
Materials	None	
Instructions	Observe if the child cries when handled.	
Response	The child may cry occasionally when handled.	
Grading	Occasional crying when handled Frequent crying when handled Cries even when not handled	

1.7.4	Sleep/awake rhythm	Age: Birth - <1 month
Materials	None	
Instructions	Ask the caregiver about the infant's sleep patterns.	
Response	Newborns sleep on average 16 to 17 hours a day, though there is rather wide variation. Cycles of sleep and waking are relatively brief. The sleep patterns may be either typical, altered sometimes, or constantly altered wherein either the infant is sleeping too much or is awake for longer periods of time.	
Grading	Normal sleep pattern Sleep awake rhythm altered sometimes Sleep awake rhythm altered constantly	

1.8 Seizure

1.8.1	Seizures	Age: Birth - <1 month
Materials	None	
Instructions	Ask caregiver or check medical records to determine whether infant has ever experienced an unprovoked seizure. Seizures are manifestations of neuronal injury due to excessive electrical activity in the brain. Any disturbance in the brain may cause seizures, which may be single or repetitive, and involves a sudden change in behaviour.	
Response	Consider typical if the infant has not had a provoked or unprovoked seizure.	
Grading	Never occurred Transient (once/repeated for one day) Repeated seizure continued for several days	